

Chapter 15: Materials We Use

Textbook Exercise Solutions & Answers



1. Fill appropriate terms in the blanks:

(white cement, soap, detergent, wearing of bones, tooth decay, hard, soft, portland, fatty acid)

(a) The substance that helps water to remove dirt from the surface of material is called

soap

(b) Fluoride is used in toothpaste to prevent

tooth decay

(c) Soap is a salt of and sodium hydroxide.

fatty acid

(d) Synthetic detergents can be used in water as well.

hard

(e) For construction purposes Cement is the most commonly used cement.

white cement

2. Write answers to the following questions.

(a) How does the use of a detergent help to clean soiled clothes?

1. A molecule of a detergent holds on to a water molecule at one end and an oil molecule at the other.
2. As a result the molecules of oil mix with the water.
3. This is how detergent acts on our soiled clothes. So detergent removes any oil or dirt sticking on to our clothes.
4. Due to the property of holding on to both oil and water, soap water spreads easily on many types of surfaces.
5. This property of spreading on a surface is called surface activity.
6. Detergents are surface active.
7. One effect of surface activity is lather formation.

(b) How will you check with the help of soap powder whether water is hard?

1. In the hard water, (of a well or a tube-well), soap does not give lather but forms a scum.
2. As a result, soap loses its cleansing property. So with help of soap we will come to know that it is a hard water.

(c) What are the important ingredients of a tooth paste and what is the function of each?

1. The important ingredients of a toothpaste are calcium carbonate and calcium hydrogen phosphate.
2. They remove the dirt on teeth. These ingredients also polish the teeth.
3. A certain proportion of fluoride in the tooth paste helps prevent tooth decay.
4. It is essential for the enamel covering of teeth.

(d) What are the ingredients of cement?

1. Cement is a dry, greenish grey powder with fine particles.
2. It is made from silica (sand), alumina (aluminium oxide), lime, iron oxide and magnesia (magnesium oxide).

(e) What will happen if cement is not used in making concrete?

1. In making concrete strong, the ingredients cement, water, sand and gravel should usually be mixed in the ratio of 1 : 2 : 3 : 0.5 to achieve maximum strength.
2. If enough cement is not used then the concrete will eventually fall apart, as cement is used as a binding agent.

(f) Make a list of detergents, that you use.

There are two types of detergents that we use, (a) Natural (b) Man-made detergents.

1. Natural detergents are soap nut (ritha) soap pod (shikekai).
2. Man-made detergents are soap, hard soaps, soft soaps.
3. Synthetic detergents have taken the place of soap.
 - Detergents are commonly available as powders or concentrated solution.
 - Detergents are for laundry, washing clothes and cleaning dishes.
4. Alkaline detergents used for hard surface cleaning.

(g) What should be expected from a detergent for delicate garments?

1. Detergents should not be strong, they may spoil the texture, colour of delicate garments.
2. Now a days many detergents are available especially for woollen, delicate clothes.
3. They should not contain bleach, phosphorous which will harm delicate clothes.

(h) What is meant by surface activity? Name three chemicals responsible for the surface activity of various detergents.

1. Due to the property of holding on to both oil and water, soap water spreads easily on many types of surfaces.
2. The property of a substance of spreading on a surface is called surface activity and the substance is called surfactant.
3. Detergents are surface active.
4. They increase the spreading and wetting ability of water by reducing its surface tension.
5. Chemicals responsible for the surface activity of various detergents is phenol, Aprotinin.

3. What are the similarities and differences between:

(a) Natural detergents and Man-made detergents

Natural detergents	Man-made detergents
1. Soap nut (ritha) and soap pod (shikekai) are the natural detergents.	1. Soap is man-made detergent soft soap and hard soap.
2. Natural detergents do not have harmful effect on human skin or on silk or woollen clothes.	2. Soap has effect on skin and cloth because it has chemicals in it.
3. Soap nut, soap pod contain a chemical saponin.	3. Soap has acid in it, soap is sodium and potassium salt of fatty acids.
4. It is naturally available.	4. It is man made using chemicals and oil.

(b) Soap and Synthetic detergents

Soap	Synthetic detergents
1. Soap is man-made detergent, it is in use since ancient times.	1. Synthetic detergent have taken the place of soap.
2. Soap loses its cleansing property in hard water it does not give lather.	2. Synthetic detergent can be used in hard water also.
3. Soap was prepared by using animal fat and wood ash.	3. Synthetic detergent are obtained by subjecting fats, kerosene to various chemical processes.
4. Variety of soaps are available, soft soap for bathing, hard soap for washing, cleaning.	4. Variety of synthetic detergents are available – Ariel, Surf Excel, etc.

(c) Bath soap and Soap for washing clothes

Bath soap	Soap for washing clothes
1. Soft soap is used for bathing.	1. Hard soap is used for washing clothes.
2. Soft soap is potassium salt of fatty acids.	2. Hard soap is sodium salt of fatty acid.
3. Soft soap dissolves more easily readily in water.	3. Hard soap do not dissolve easily in water.
4. Now a days many perfumes, moisturizers are mixed to make it more attractive and useful.	4. In this type of soap now a days germicides, anti foaming agents are mixed to make it more useful.

(d) Modern cement and Ancient cement

Modern cement	Ancient cement
1. Cement manufactured from 60% lime, silica 25% and alumina 5%. the rest is iron oxide and gypsum.	1. Romans used to make aqueous cement by mixing volcanic ash in moistened lime. It was very durable.
2. It is the most commonly used cement for Construction work.	2. It is Roman cement volcanic ash prevent cracks from spreading.
3. It is called Portland Cement.	3. It is called Aqueous Cement.

4. Explain why –

(a) Soap cannot be used in hard water.

1. In the hard water of a well or a tube well, soap does not give lather but forms a scum.
2. As a result soap loses its cleansing property. So soap cannot be used in hard water.

(b) Oil does not mix in water. However, oil and water become homogenous if a sufficient quantity of detergent is added.

1. A molecule of a detergent holds on to a water molecule at one end and an oil molecule at the other.
2. As a result the molecules of oil mix with the water, and we will see that the water and oil have become homogenous and the colour of the mixture appears milky.

(c) Synthetic detergents are superior to soap.

1. Soap is a man-made detergent which has been in use since ancient times.
2. It was prepared by using animal fat and wood ash. In hard water, soap does not give lather, it loses its cleansing property.
3. So we can not use soap in hard water. Now synthetic detergents have taken the place of soap.
4. They can be used in hard water as well
5. Synthetic detergents have many more advanced properties like they are added with perfumes, conditioner for fabric, etc.

(d) Often coloured spots are formed on clothes during washing.

1. Curry stains stick fast to the criss-crossing threads of the material of our clothes.
2. Curry contains turmeric a natural indicator which turns red in a basic solution.
3. Soaps are sodium salts of fatty acids and are basic in nature.
4. While washing the clothes, turmeric reacts with soap solution and turns red forming red spots on the clothes.
5. But the spots disappear after washing the clothes with plenty of water.

(e) Tobacco masheri should not be used for cleaning teeth.

1. Masheris is the smokeless form of tobacco. It is tobacco, containing teeth cleaning powder
2. It contains tobacco leaves.
3. Tobacco contains toxic and Carcinogenic chemicals which can cause cancer, oral cancer, mouth and throat cancer, gum disease, tooth decay.
4. Its use can cause abnormal delivery in pregnant women.